MOHAMED AHMED, Ph.D.

Lead Data Scientist | Adjunct Professor

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Professional Profile

Geographer, geospatial data scientist, and educator with more than 15 years of experience applying GIS, remote sensing, machine learning, and ocean data science to environmental, educational, and public-interest challenges. My current work lies at the intersection of geography, GIS, and applied geospatial research, with a particular focus on GeoAI, remote sensing, Arctic and marine carbon systems, big data analytics, and the translation of the cutting-edge geospatial methods and techniques into usable tools, workflows, and learning resources.

As Lead Data Scientist and GeoAI Lead with Esri Canada's Education and Research group, I support academic, government, and research organizations across Canada in adopting GeoAI and advanced geospatial analytics. I also serve as an Adjunct Professor in Geography Department at the University of Calgary, where I teach and mentor students in GIS, remote sensing, and oceanography.

My research and professional contributions include peer-reviewed publications on the Canadian Arctic, ocean acidification, remote sensing, and environmental data science; leadership in international research partnerships; delivery of invited guest lectures, workshops, and conference presentations; and development of public-facing technical writing and knowledge-mobilization resources. Across these roles, my work is grounded in the belief that geographic knowledge, when combined with responsible technology and education, can help Canadians better understand Canada's landscapes, environments, communities, and future challenges.

Core Areas of Contribution

Geography & GeoAI	Leadership in applying geospatial AI, remote sensing, computer vision, spatial data science, and GIS to environmental, educational, and operational challenges.
Canadian Relevance	Research and applied partnerships focused on the Canadian marine and terrestrial ecosystems, with focus on forest and marine carbon fluxes, ocean acidification, freshwater biodiversity, wildfire evacuation, and northern ecosystem stewardship.
Education & Mentorship	Teaching, technical training, workshops, invited lectures, graduate-student support, and mentorship in GIS, remote sensing, GeoAI, and environmental data science.
Knowledge Mobilization	Technical blogs, StoryMaps, public-facing learning resources, conference demos, and research translation for academic and non-technical audiences.

Selected Contributions to Geographic Education, Research, And Public Engagement

- Delivered more than 50 invited lectures, technical workshops, and conference presentations across Canadian universities, government agencies, and national and international events.
- Authored +10 technical blogs and learning resources on GeoAI, deep learning, biodiversity monitoring, climate change, and emerging geospatial technologies.
- Teach undergraduate and graduate students in geography, GIS, remote sensing, spatial data science, and applied geospatial technologies as an Adjunct Professor at the University of Calgary.
- Serve as a Program Advisory Committee member for SAIT's Bachelor of Applied Technology Geographic Information Systems (BGIS) program, advising on curriculum alignment with industry standards, technological change, and GIS employment requirements.
- Mentor students, research assistants, and early-career researchers in geospatial analysis, remote sensing, environmental data science, and career development in geography.
- Supported interdisciplinary ocean data science education through the Pacific Rim Ocean Data Mobilization and Technology program, contributing to training and curriculum activities across universities in Canada.
- Participated in academic service as a peer reviewer for journals in geoscience, remote sensing, biogeochemistry, oceanography, and environmental science.

Selected Academic Research Partnerships and Applied Contributions at Esri Canada

Marine Mammal Monitoring using AI

Collaborators: Fisheries and Oceans Canada

- Supported applied GeoAI work for marine mammal monitoring in Canadian waters, including development and communication of deep learning workflows for detecting beluga whales.
- The Beluga Whale detection using AI was recognized among Esri's 10 most-read stories of 2025.
- Currently supporting a new master's student at Fisheries and Oceans Canada who is fine-tuning customized deep learning models to improve detection of other marine mammals across Canadian waters.

Canada Ocean Acidification Mapping and Knowledge Mobilization

Collaborators: Fisheries and Oceans Canada, MEOPAR, University of Calgary, Hakai Institute, Memorial University, Dalhousie University, University of Victoria

- Contributed to a national ocean acidification mapping and knowledge-mobilization initiative focused on improving access to Canadian ocean acidification data, research assets, and decision-support resources.
- Co-authored the peer-reviewed paper Ocean acidification in Canada: the current state of knowledge and pathways for action, published in Frontiers in Marine Science.
- Supported the project's emphasis on connecting scientific evidence, geospatial analysis, data sharing, and policy-relevant decision-making to better address ocean acidification across Canada.

Detecting Invasive Species and Mammal Communities Using iDNA and GIS

Collaborators: University of Guelph

- Support a Mitacs MSc research project exploring how leech-derived invertebrate DNA can be used as a scalable, non-invasive method for monitoring mammal communities in Canadian freshwater ecosystems.
- The project combines field sampling from rivers and wetlands in Ontario, DNA metabarcoding, spatial analysis, remote sensing, and the use of environmental data from ArcGIS Living Atlas of the World.

Adaptive Route Optimization for Wildfire Evacuation

Collaborators: University of Calgary

- Support a Mitacs PhD project developing an AI-powered adaptive route optimization system for wildfire evacuation, using the 2025 Flin Flon wildfire in Saskatchewan and Manitoba as a case study.
- The project integrates GPS movement data, reinforcement learning, GIS, and interactive dashboard design to support safer evacuation planning for evacuees and emergency managers; the student presented the work at a University of Calgary students conference and received a best paper award.

Creeburn CLEAR Hub: Indigenous Knowledge, Technology, and Ecosystem Restoration

Collaborators: University of Calgary, Fort McKay First Nation

- Support the work for the Creeburn CLEAR Hub, a living laboratory in Northern Alberta focused on combining Indigenous knowledge, ecosystem monitoring, land restoration, and sustainable off-grid solutions.
- The initiative aims to support ecosystem stewardship, climate goals, green jobs, and scalable approaches for sustainable land management in regions affected by industrial activity.

Spatial Aggregation and Big Data Analytics

Collaborators: University of New Brunswick

- Support a Mitacs PhD research project advancing the integration of geospatial machine learning models as relational algebra operators within standard SQL workflows.
- The research has potential to make advanced geospatial analytics more accessible within existing data systems and to strengthen decision-making using real-time and near-real time big datasets.

Accurate Forest Carbon Quantification

Collaborators: York University

- Support ongoing research with York University on forest carbon quantification and the potential application of Esri technology to improve geospatial workflows, data integration, and applied analysis.
- The project is relevant to Canadian forest carbon monitoring, nature-based climate solutions, and the development of more accurate methods for assessing forest carbon dynamics.

Pacific Rim Ocean Data Mobilization and Technology Program

Collaborators: University of British Columbia, University of Victoria, University of Waterloo

- Contributed to an interdisciplinary training initiative focused on ocean observation, prediction, data analysis, numerical modelling, forecasting, and knowledge mobilization.
- Supported development of job-ready ocean data science skills for students and early-career researchers working across ocean technology, fisheries, aquaculture, geophysics, government, academia, and NGOs.

Two-Eyed Seeing for Forest Vegetation, Integrity, and Nature-Based Climate Solutions

Collaborators: York University

- Support a five-year NSERC Alliance project focused on accurate quantification of forest vegetation and integrity for nature-based climate solutions.
- The project applies AI-driven approaches while incorporating Two-Eyed Seeing and Indigenous knowledge systems to improve understanding of forest integrity, biodiversity, and ecosystem condition.

PROFESSIONAL EXPERIENCE

Lead Data Scientist & GeoAI Lead

2022 – Present

Esri Canada | Education and Research Group

- Lead GeoAI initiatives within Esri Canada’s Education and Research group, advancing the use of AI, machine learning, deep learning, and spatial data science for academic, government, and research organizations across Canada.
- Develop applied geospatial AI workflows for computer vision, object detection, image classification, change detection, land-use and land-cover analysis, and large-scale spatial data processing.
- Translate emerging GeoAI capabilities into practical workflows, training resources, and decision-support approaches that help researchers and organizations apply geographic knowledge to real-world challenges.
- Author technical blogs, learning resources, and knowledge-mobilization materials that make advanced geospatial AI methods more accessible to technical and non-technical audiences.
- Deliver guest lectures, technical workshops, and conference presentations across Canadian universities, government agencies, and national and international events.

Adjunct Professor

2025 – Present

University of Calgary, Canada

- Teach undergraduate and graduate students in the Department of Geography, with a focus on GIS, remote sensing, and applied geospatial technologies.
- Mentor students on research projects, technical skill development, and career pathways in geography, with a focus on applying geospatial methods to real-world environmental and societal challenges.
- Connect classroom learning with industry practice by introducing applied case studies, emerging geospatial technologies, and real-world workflows.
- Support the next generation of geographers and geospatial professionals by emphasizing spatial thinking, practical problem-solving, and responsible use of geospatial data.

Senior Researcher & Project Manager

2022 – 2025

eLEAF B.V., Netherlands (Part-Time)

- Led an international applied research project across Africa and Europe focused on estimating land-based negative carbon emissions using remote sensing, biogeochemical modelling, and geospatial data systems.
- Coordinated a multidisciplinary team to design, validate, and document a prototype data system for environmental monitoring and carbon-cycle assessment.
- Produced and co-authored five technical deliverables for EU Horizon Europe, covering data architecture, calibration, validation, implementation workflows, and applied research translation.
- Worked with government, NGO, research, and community stakeholders to align technical development with policy needs, local constraints, ethical considerations, and long-term scalability.

Postdoctoral Researcher
2021 – 2022

University of Victoria, Canada

- Designed and implemented machine learning models to analyze large-scale oceanographic observations from the global Argo float network, supporting new estimates of oxygen variability in the Northwest Pacific.
- Built reproducible workflows for Argo float profile data, including data acquisition, quality control, profile preparation, statistical analysis, and interpretation.
- Coordinated the PRODIGY interdisciplinary ocean data science program across four universities in Canada and Chile, supporting curriculum design, delivery, and training in applied ocean data science.

Postdoctoral Researcher
2020 – 2021

University of Calgary, Canada

- Designed and implemented a GIS and machine learning data pipeline integrating time-series satellite imagery, in-situ Arctic field observations, and neural network models to estimate Canada's marine carbon sink and support national climate reporting in the Arctic.
- Supervised three research assistants by assigning technical tasks, providing analytical guidance, monitoring progress, and maintaining data quality and project timelines.
- Conducted biogeochemical field and laboratory experiments across Arctic ship expeditions, integrating multi-source observations into a unified analytical framework.

Graduate Research Scientist
2015 – 2020

University of Calgary, Canada

- Engineered a scalable geospatial ML workflow combining satellite imagery, GIS, and ship-based field observations to estimate Arctic Ocean carbon uptake from multi-year field expeditions.
- Designed data quality, metadata, and standardization pipelines to transform raw underway sensor datasets into validated, analysis-ready, and publicly archived data products through NOAA.
- Served as a core data scientist in multi-institutional research consortia, coordinating data sharing, analytical workflows, and integration protocols across universities, government agencies, and international partners.

Research Associate
2012 – 2013

Lund University, Sweden

- Developed an automated GIS and remote sensing workflow to process raw satellite imagery into standardized analytical outputs for a government-funded land cover dynamics project across Egypt.
- Applied geospatial analysis, image processing, and spatial data interpretation to assess land cover change and climate-related environmental impacts.
- Mentored two graduate students on geospatial analysis and vegetation dynamics across the African Sahel.

Assistant Lecturer
2013 – 2015

Beni-Suef University, Egypt

- Designed and taught undergraduate and graduate courses in GIS, remote sensing, hydrogeology, aerial photography, and applied spatial data analysis.
- Led multi-week geological field campaigns for groups of 20 students across Egypt's Red Sea coast and Sinai Peninsula, supporting field-based training, data collection, and applied interpretation.
- Supervised eight undergraduate students using remote sensing for earth science applications.

EDUCATION

Ph.D., Geography (GPA: 4.0) | University of Calgary, Canada — 2015 – 2020

Thesis: Air-Sea CO_2 Cycling in Arctic Coastal Seas: Case Studies in the Canadian Arctic Archipelago and Hudson Bay

M.Sc. (with Distinction), Geomatics | Lund University, Sweden — 2010 – 2012

Thesis: Significance of Soil Moisture on Vegetation Dynamics in the African Sahel, 1982–2008

B.Sc. (Honours, 1st Class), Geology | Beni-Suef University, Egypt — 2003 – 2007

Thesis: Environmental pollution of cement industry: A case study in Beni-Suef city, Egypt.

PEER-REVIEWED PUBLICATIONS

Lead-Authored:

1. **Ahmed, M.**, Prikaziuk, E., Laub, M., Klaasse, A.L., Ellerbroek, L.E. (2025). A novel approach to mapping and monitoring land carbon sinks by combining remote sensing and biogeochemical modeling: A case study in Burkina Faso. Ecological Informatics, 90, 103174.
2. **Ahmed, M.**, Else, B.G.T., Butterworth, B., Capelle, D.W., Gueguen, C., Miller, L.A., Meilleur, C., Papakyriakou, T. (2021). Widespread surface water pCO_2 undersaturation during ice-melt season in an Arctic continental shelf sea (Hudson Bay, Canada). Elementa: Science of the Anthropocene, 9(1): 00130.
3. **Ahmed, M.**, Else, B.G.T., Capelle, D., Miller, L.A., Papakyriakou, T. (2020). Underestimation of surface pCO_2 and air-sea CO_2 fluxes due to freshwater stratification in an Arctic shelf sea, Hudson Bay. Elementa: Science of the Anthropocene, 8(1): 084.
4. **Ahmed, M.**, Else, B. (2019). The Ocean CO_2 sink in the Canadian Arctic Archipelago: A present day budget and past trends due to climate change. Geophysical Research Letters, 46, 9777–9785.
5. **Ahmed, M.**, Else, B., Burgers, T., Papakyriakou, T. (2019). Variability of surface water pCO_2 in the Canadian Arctic Archipelago from 2010 to 2016. Journal of Geophysical Research: Oceans, 124(3), 1876–1896.
6. **Ahmed, M.**, Else, B., Eklundh, L., Ardö, J., Seaquist, J. (2017). Dynamic response of NDVI to soil moisture variations during different hydrological regimes in the Sahel region. International Journal of Remote Sensing, 38(19), 5408–5429.

Co-Authored:

7. Deschepper, I., Stefels, J., Corkill, M., **Ahmed, M.**, & Delille, B. (2026). Sea ice biogeochemistry. In Reference Module in Earth Systems and Environmental Sciences. Elsevier. https://doi.org/10.1016/B978-0-323-85242-5.00076-2
8. Barclay, K.M., Gurney-Smith, H.J., **Ahmed, M.**, et al. (2026). Ocean acidification in Canada: the current state of knowledge and pathways for action. Frontiers in Marine Science, 13, 1761703.
9. Westbrook, H.C., Bourbonnais, A., Manning, C.C.M., Tremblay, J.-É., **Ahmed, M.**, Else, B., Granger, J. (2024). Dissolved nitrogen cycling in the Eastern Canadian Arctic Archipelago and Baffin Bay from stable isotopic data. Global Biogeochemical Cycles, 38(12), e2023GB007926.
10. Nickoloff, G., Else, B.G.T., **Ahmed, M.**, Burgers, T.M., Miller, L.A., Papakyriakou, T. (2024). Strong pCO_2 undersaturation in an Arctic Sea: A decade of spatial and temporal variability in Baffin Bay. Journal of Geophysical Research: Oceans, 129(8), e2023JC020595.
11. Duke, P.J., Hamme, R.C., Ianson, D., Landschützer, P., **Ahmed, M.**, Swart, N.C., Covert, P.A. (2023). Estimating marine carbon uptake in the northeast Pacific using a neural network approach. Biogeosciences, 20(18), 3919–3941.
12. Duke, P.J., Richaud, B., Arruda, R., Länger, J., Schuler, K., Gooya, P., **Ahmed, M.**, et al. (2023). Canada's marine carbon sink: an early career perspective on the state of research and existing knowledge gaps. Facets, 8, 1–21.
13. Sims, R.P., **Ahmed, M.**, Butterworth, B.J., Duke, P.J., Gonski, S.F., Jones, S.F., Brown, K.A., Mundy, C.J., Williams, W.J., Else, B.G.T. (2023). High interannual surface pCO_2 variability in the southern Canadian Arctic Archipelago's Kitikmeot Sea. Ocean Science, 19(3), 837–856.
14. Falardeau, M., Bennett, E.M., Else, B., Fisk, A., Mundy, C.J., Choy, E.S., **Ahmed, M.**, Harris, L.N., Moore, J.-S. (2022). Biophysical indicators and Indigenous and Local Knowledge reveal climatic and ecological shifts with implications for Arctic Char fisheries. Global Environmental Change, 74, 102469.
15. Barber, D.G., Harasyn, M.L., Babb, D.G., Capelle, D., et al., **Ahmed, M.**, Else, B., et al. (2021). Sediment-laden sea ice in southern Hudson Bay: Entrainment, transport, and biogeochemical implications. Elementa: Science of the Anthropocene, 9(1): 00108.
16. Duke, P.J., Else, B.G.T., Jones, S.F., Marriot, S., **Ahmed, M.**, Nandan, V., et al. (2021). Seasonal marine carbon system processes in an Arctic coastal landfast sea ice environment observed with an innovative underwater sensor platform. Elementa: Science of the Anthropocene, 9(1): 00103.
17. Manning, C.C., Preston, V.L., Jones, S.F., Michel, A.P.M., Nicholson, D.P., Duke, P.J., **Ahmed, M.**, Manganini, K., Else, B.G.T., Tortell, P.D. (2020). River inflow dominates methane emissions in an Arctic coastal system. Geophysical Research Letters, 47(10), e2020GL087669.
18. Harris, L.N., Yurkowski, D.J., Gilbert, M.J.H., Else, B.G.T., Duke, P.J., **Ahmed, M.**, Tallman, R.F., Fisk, A.T., Moore, J.-S. (2020). Depth and temperature preference of anadromous Arctic char in the Kitikmeot Sea, a shallow and low-salinity area of the Canadian Arctic. Marine Ecology Progress Series, 634, 175–197.

TECHNICAL SKILLS

GeoAI & Computer Vision	CNN-based architectures, object detection, image classification, semantic segmentation, change detection and land cover/land use classification, geospatial foundation-model workflows, and model evaluation.
Spatial Data Science & Geospatial Platforms	Spatial-temporal analytics, raster analytics, satellite imagery analysis, enterprise GIS, ArcGIS Pro, ArcGIS Online, ArcGIS Notebooks, ArcGIS Python API, Google Earth Engine, ENVI, and QGIS.
Data Engineering & Reproducibility	Python, SQL, Shell Scripting, QA/QC pipeline design, metadata standards, open data publishing, data lifecycle management, and reproducible analytical workflows.
Scientific Computing & Analytics	MATLAB, R, SPSS, statistical modelling, exploratory data analysis, time-series analysis, scientific visualization, and oceanographic/environmental data analysis.
Development Tools & Systems	Linux, Windows, Git, notebook-based analytics, technical documentation, and cloud-integrated workflows.

SELECTED AWARDS & RECOGNITION

- PRODIGY Fellow — Pacific Rim Ocean Data Mobilization & Technology Program ($70,000, 2021)
- Geography Department Excellent Research Award — University of Calgary (2020)
- Dean's Doctoral Scholarship — University of Calgary ($15,000, 2019)
- Student Educational Fellowship — University of Manitoba ($10,000, 2019)
- Esri Canada GIS Scholarship ($2,500, 2017)
- Maritime Awards Society of Canada Graduate Scholarship ($2,500, 2017)
- Best Oral Presentation Award — Annual Geography Department Conference, University of Calgary (2018)
- Eyes High International Doctoral Scholarship — University of Calgary ($12,000, 2016)
- John W. Davies Memorial Award — University of Calgary ($3,000, 2016)
- ArcticNet Student Training Fund Awards ($5,000 in 2016; $2,790 in 2018)
- EU Bridge Building Scholarship — Lund University, Sweden ($35,000, 2010–2012)
- Top Student Fellowship — Beni-Suef University ($15,000, 2009)
- Faculty Excellence Student Award — Beni-Suef University (2008)

PROFESSIONAL AFFILIATIONS & ACADEMIC SERVICE

- Program Advisory Committee Member, Bachelor of Applied Technology Geographic Information Systems (BGIS), Southern Alberta Institute of Technology (SAIT) (2025–Present): Advise on curriculum alignment with industry standards, emerging geospatial technologies, and employment needs to help prepare students for successful GIS careers.
- Peer Reviewer (2018–present): Geophysical Research Letters, Remote Sensing of Environment, Global Biogeochemical Cycles, Frontiers in Marine Science, Elementa: Science of the Anthropocene, Geo-spatial Information Science, Limnology & Oceanography Letters
- Member: American Geophysical Union, European Geosciences Union, Canadian Meteorological and Oceanographic Society, Global Ocean Acidification Observing Network (GOA-ON), Canada's Ocean Acidification Community of Practice, ArcticNet, Association of Polar Early Career Scientists, Surface Ocean–Lower Atmosphere Study (SOLAS), NetCOLOR

LEADERSHIP & COMMUNITY CONTRIBUTIONS

- **President, Egyptian Student Association**, University of Calgary (2016–2017): Led the Executive Council, coordinated social and academic events, and managed association communications through social media and campus outreach.
- **Chair, Finance Committee**, The Graduate College, University of Calgary (2017–2018): Developed financial policies and budgets for student committee operations; raised more than $3,000 through fundraising.

- **Service Committee Lead**, The Graduate College, UCalgary (2020–2022): Led volunteer initiatives supporting immigrants and homeless communities; coordinated river and pathway cleanup programs.
- **Vice President**, Student Life, Geography Department, UCalgary (2016–2018): Organized academic and social programming, including a weekly lunch-and-talk lecture series.
- **My GradSkills Leader Ambassador**, University of Calgary (2015–2019): Connected graduate students with professional development resources, training opportunities, and campus supports to help them succeed in their programs.
- **Vice President, Student Life**, Students Union, Beni-Suef University (2005–2007): Planned social and sports activities for students; advocated for student well-being and mental health awareness across campus.

Selected Technical Writing & Knowledge Mobilization

- Ahmed, M. (2026). What Makes a Place Runnable? Q&A with SKI Best Paper Winner
- Steven Edwards & Mohamed Ahmed (2026): Teaching and Learning GIS in the Age of AI (Part 1 of 3)
- Ahmed, M. (2025). Building Damage Assessment with GeoAI in ArcGIS
- Ahmed, M. (2025). Arctic Sea Ice Decline (1979-2024)
- Ahmed, M. (2025). Building Footprints Extraction using Deep Learning
- Ahmed, M. (2025). Automated Road Extraction Using Deep Learning
- Ahmed, M. (2024). Hotter Oceans
- Ahmed, M. (2024). Addressing and Understanding Climate Change with GIS
- Ahmed, M. (2024). eDNA and GIS for Powerful and Scalable Biodiversity Monitoring – Part 2
- Ahmed, M. (2024). eDNA and GIS for Powerful and Scalable Biodiversity Monitoring – Part 1
- Ahmed, M. (2024). Visualizing Tipping Elements to Better Understand Climate Change – Part 3
- Ahmed, M. (2023). Visualizing Tipping Elements to Better Understand Climate Change – Part 2
- Ahmed, M. (2023). Visualizing Tipping Elements to Better Understand Climate Change – Part 1
- Ahmed, M. (2023). Importing and Visualizing Near-Real Time Argo Floats Data in ArcGIS.
- Ahmed, M. (2023). Pretrained Deep Learning Models: Why Are They Useful and How to Use Them?
- Ahmed, M. (2023). Unlocking the Potential of Deep Learning in the ArcGIS System.

REFERENCES

Available upon request.